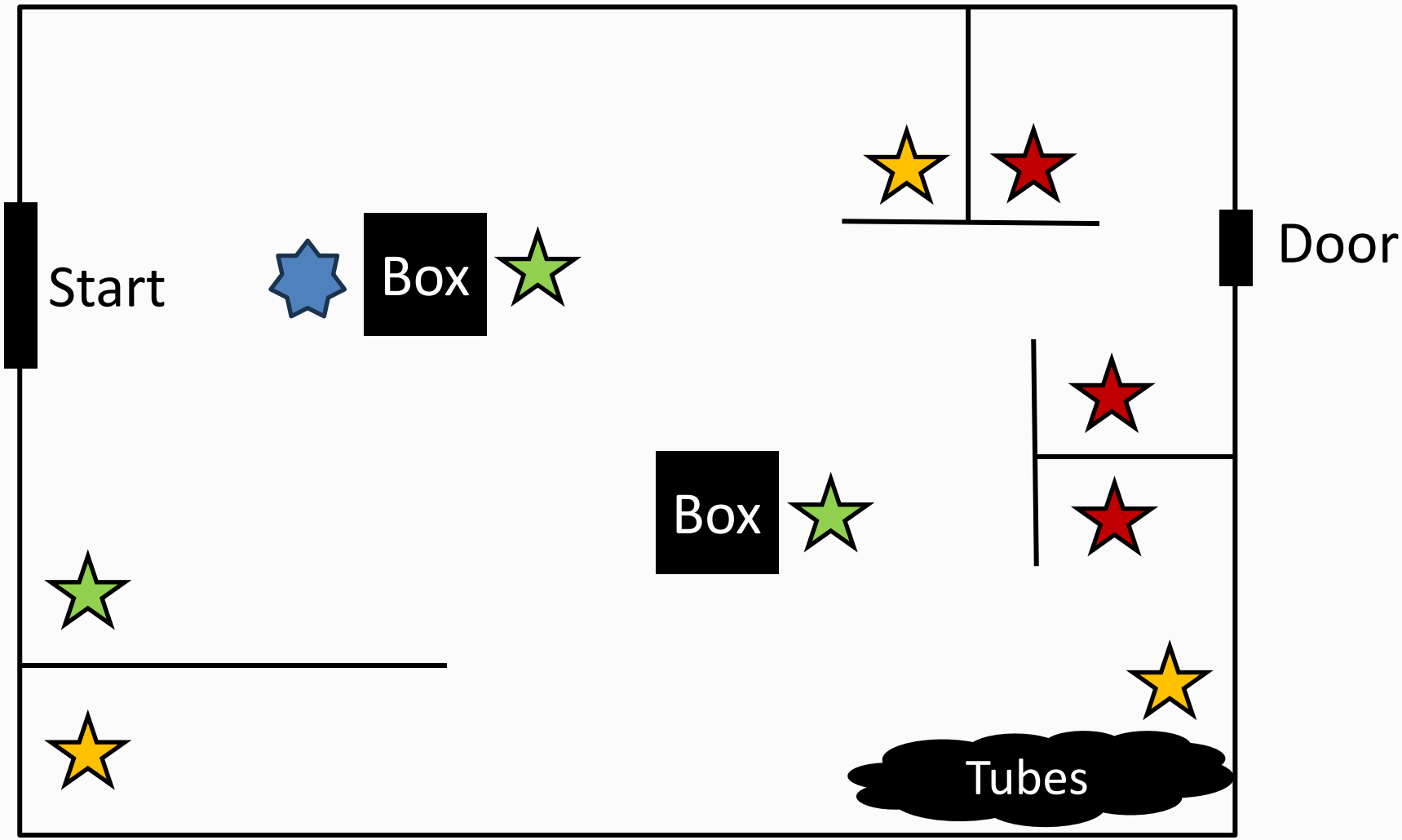
Venue

Easy 2/3 + Middle 2/3 + Difficult 2/3 = 6/9position



6/10 Object



Clock



Teddy Bear



Bowl



Apple



Banna



Vase



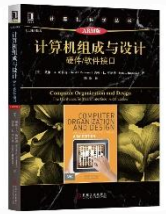
Cup



Cake



Bottle



Book



Supplement Information

1. Objects will only be placed at the nine predefined locations.

2. Each of the three task levels uses a different scene. After completing a level, the robot must return to the starting point to receive the tasks for the next level. The scene and corresponding map will also be changed.

3. Each scene contains six objects selected from nine available object categories. One object is placed at each of six locations: two Easy, two Medium, and two Difficult.

1. 物体仅会摆放在预先设定的9个固定点位中。

2. 三个任务等级（Level 1–3）分别使用不同的场景。完成一个等级的全部任务后，机器人需返回起始点，领取下一等级的任务，同时更换场景及对应地图。

3. 每个场景摆放6个物体，物体类别从9类可选类别中选取。6个物体分别放置在6个点位上，其中简单（Easy）、中等（Medium）和困难（Difficult）点位各2个。

Supplement Information

4. For the two Level 1 tasks, one target location will be randomly selected from the two Easy locations, and the other from the two Medium locations.

5. For the two Level 2 tasks, one target location will be randomly selected from the two Easy locations, and the other from the two Difficult locations.

4. Level 1包含两个任务：从2个简单点位和2个中等点位中，分别随机选择1个作为任务目标点位。

5. Level 2包含两个任务：从2个简单点位和2个困难点位中，分别随机选择1个作为任务目标点位。

Supplement Information

6. You may use any large language model API of your choice. The course provides a reimbursement allowance of RMB 100, with reimbursements to be processed together at a later date. We recommend using these APIs for text recognition and interpreting task instructions in Level 3.

7. A mounting bracket is required to attach the Stage 1 camera to the robot. Please contact a teaching assistant for installation.

6. 大语言模型API可自由选择。课程提供100元报销额度，后续统一办理报销。建议将API用于文字识别及Level 3的任务指令理解。

7. 将Stage 1使用的相机集成至机器人时，需要安装支架，请联系助教协助安装。

* Grading criteria

Stage 1 Stereo Vision: 40% in total

1. Binocular camera preparation and calibration: 25%
2. Stereo matching performance metrics: 15%

accuracy 5%, speed 5%, compactness 5%

Details:

1. Obtain the calibration matrix 25%

Place an object within the field of view of the stereo camera (approximately 20-50 cm from the camera), and output the camera's depth map.

1. Depth map is complete with 0/<5/<10 defects 5%/3%/1%
2. The output time is within 1s/3s/5s 5%/3%/1%
3. The depth error is less than 3cm/5cm/10cm 5%/3%/1%

* Grading criteria

Stage 2 Intelligent Robot: 60% in total

1. **Detection** & **Measurement** & caption accuracy: 45%
2. Object seeking speed: 10% (ranked by overall completion time)
3. Collision avoidance ability: 5%

Details:

Detection & measurement & caption accuracy: 45% (Two Level 1 tasks at 7 points each, two Level 2 tasks at 9 points each, and one Level 3 task at 13 points.

Bonus:

1.Integrate your custom-built stereo camera into stage 2 instead of using the robot's built-in one: 10%

2. Fully autonomous operation without manual intervention: 10% Two points will be deducted for each manual intervention

* Grading criteria

Stage 1 Stereo Vision: 40% in total

1. Binocular camera preparation and calibration: 25%
2. Stereo matching performance metrics: 15%

accuracy 5%, speed 5%, compactness 5%

Stage 2 Intelligent Robot: 60% in total

1. **Detection** & **Measurement** & caption accuracy: 45%
2. seeking speed: 10% (ranked by overall completion time)
3. Collision avoidance ability: 5%

Bonus:

1.Integrate your custom-built stereo camera into stage 2 instead of using the robot's built-in one: 10%

2. Fully autonomous operation without manual intervention: 10%

Two points will be deducted for each manual intervention

Sample Program

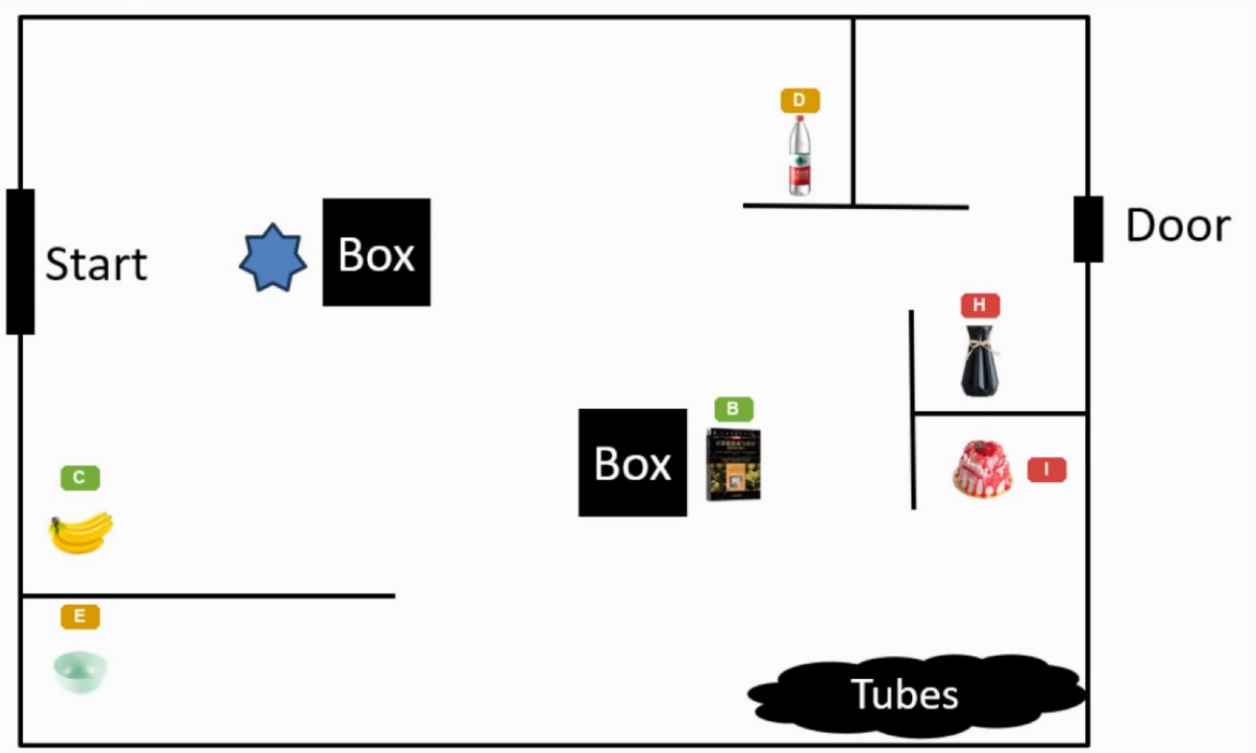
场景与任务生成器

每关物体 6 个 · 每种难度 2 个 重新开始

Level 1 / Level 2 / Level 3

Level 1 • 场景地图

● Easy • 2 个 ● Middle • 2 个 ● Difficult • 2 个



字母表示固定点位；物体旁标签的颜色表示位置难度。

先观察地图，再点击右侧按钮进入本关任务。

抽取并查看任务

场景与任务生成器

每关物体 6 个 · 每种难度 2 个 重新开始

Level 1 / Level 2 / Level 3

Level 1

从已摆放的 Easy 和 Middle 物体中各随机抽取 1 个

1. Please go to **Point B** and identify the object displayed there.

请前往 **B 点**，识别该位置的物体。

2. Please go to **Point E** and identify the object displayed there.

请前往 **E 点**，识别该位置的物体。

本关任务已固定，返回地图不会重新抽取。

返回地图 进入 Level 2

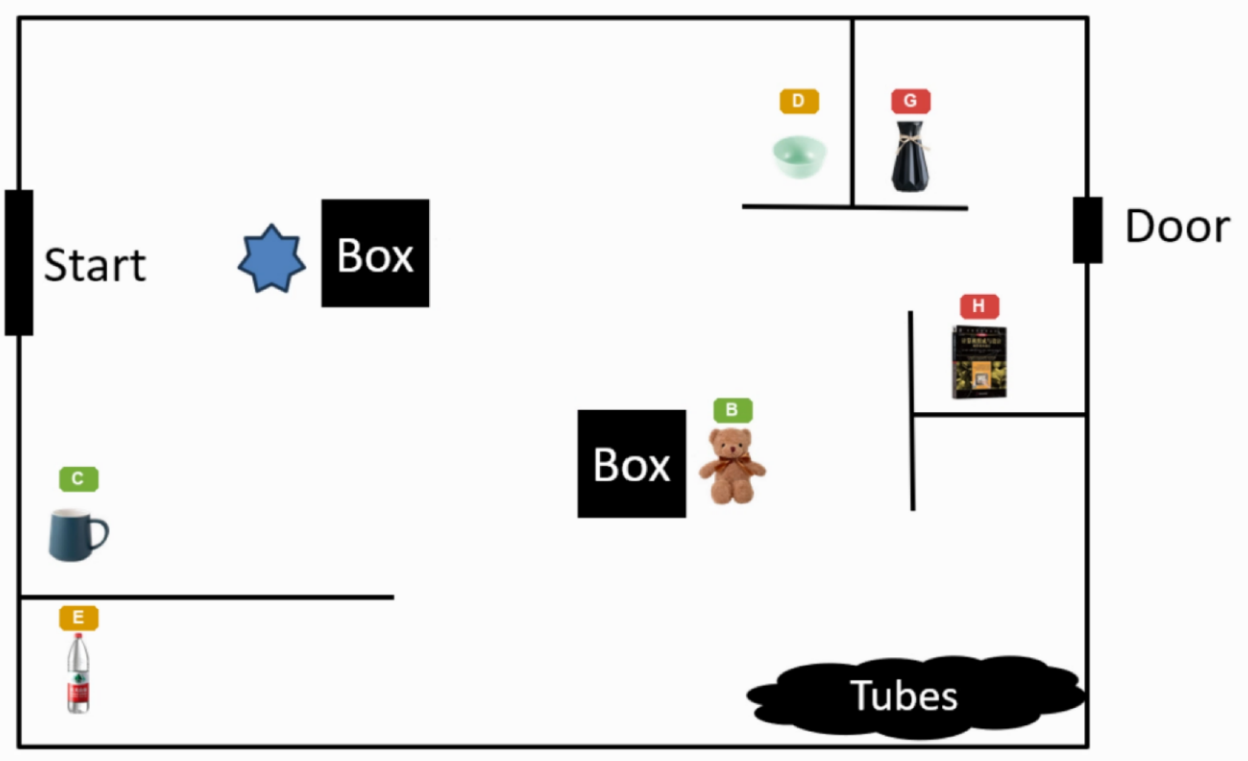
场景与任务生成器

每关物体 6 个 · 每种难度 2 个 重新开始

Level 1 ✓ / Level 2 / Level 3

Level 2 • 场景地图

● Easy • 2 个 ● Middle • 2 个 ● Difficult • 2 个



字母表示固定点位；物体旁标签的颜色表示位置难度。

先观察地图，再点击右侧按钮进入本关任务。

抽取并查看任务

场景与任务生成器

每关物体 6 个 · 每种难度 2 个 重新开始

Level 1 ✓ / Level 2 / Level 3

Level 2

从已摆放的 Easy 和 Difficult 物体中各随机抽取 1 个

1. Please find the point where the **Mug** is located.

请找出**杯子**所在的点位。

2. Please find the point where the **Vase** is located.

请找出**花瓶**所在的点位。

本关任务已固定，返回地图不会重新抽取。

返回地图 进入 Level 3

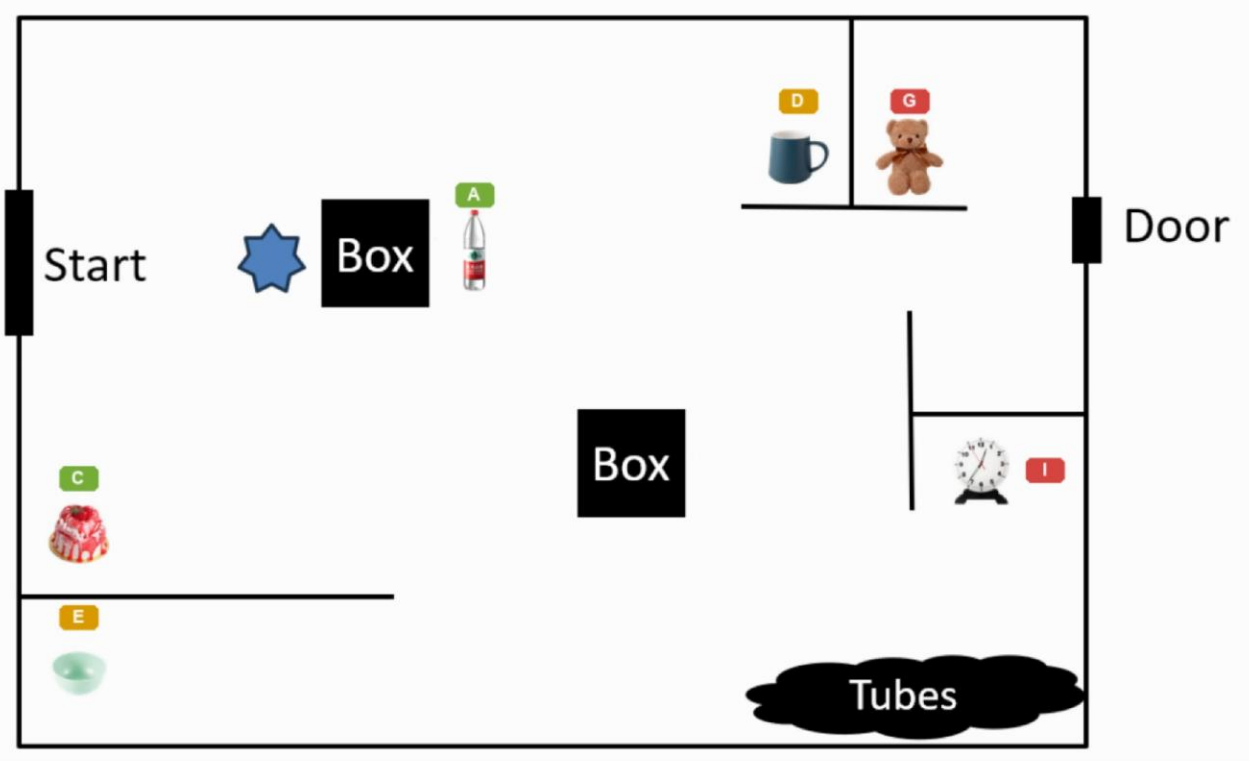
场景与任务生成器

每关物体 6 个 · 每种难度 2 个 重新开始

Level 1 ✓ / Level 2 ✓ / Level 3

Level 3 • 场景地图

● Easy • 2 个 ● Middle • 2 个 ● Difficult • 2 个



字母表示固定点位；物体旁标签的颜色表示位置难度。

先观察地图，再点击右侧按钮进入本关任务。

抽取并查看任务

场景与任务生成器

每关物体 6 个 · 每种难度 2 个 重新开始

Level 1 ✓ / Level 2 ✓ / Level 3

Level 3

从 3 种任务类型中随机抽取 1 个任务

1. Please count how many objects in the venue have the **same color** as the object at **Point D**.

请统计场景中与 **D 点**物体**主体颜色相同**的物体数量（包含该物体本身）。

本关任务已固定，返回地图不会重新抽取。

返回地图 完成本轮